Chapter 2
Metaverse Stories



Abstract This chapter situates the metaverse in a long line of utopian and dystopian visions of a digital future in which being simultaneously 'here and elsewhere' is tied to claims about disruption and revolution. It does this by tracing the main narratives about virtual worlds and their historical origins—including their escapist visions of disembodiment and spatiality—back to twentieth- and early twenty-first century scientific and fiction literature, sci-fi movies, and corporate and political documents.

2.1 To Be Here and Elsewhere

The 'Metaverse' is only the most recent term to describe a long series of dreams, desires, and fantasies that have been shaping our world from the very beginning of human history. Media and communication scholars from the Toronto School (Innis, 1950; McLuhan, 1964) up to more recent contributions such as those by Bolter and Grusin (2000), Sherry Turkle (1995) and Boyd (2014), have repeatedly framed communication technologies as 'extensions' of our individual and social reality. History itself, as an academic discipline, began with writing, a technology that extends our thoughts and utterances on paper. As much as modernity is characterized by a "proliferation of infrastructures" (Peters, 2015, 31), it is also characterized by a proliferation of media and, concurrently, by technologies, devices, languages, and styles that allow us to project our lives, ideals, and imaginations in many different ways. In this context:

> VR and AR are among the latest additions to our complex media culture, which still includes everything from books and magazines to paintings, radio, television, and film — all competing for attention and status. In that competition, the producers and promoters of each form promise an experience that is unique and, in some way, better than other media — more compelling, more authentic, truer to life. (Bolter et al., 2021, 22)

Understanding the metaverse from a historical perspective means digging into the multiple processes—technological, political, and economic but also cultural—that have led up to its current conceptualization and development. Most historical accounts of the metaverse begin with the sci-fi novel *Snow Crash* written by Neil

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Stephenson, which inspired the rebranding of Facebook to Meta (Seebold & Nam, 2025). However, authors such as Bolter et al. (2021) trace the development of virtual and augmented reality back through the history of visual culture by focusing on the various technical and aesthetic features that were step by step developed and integrated into previous media technologies, thus exemplifying the process of remediation. Similarly, Evans (2019) stresses, in his book on the re-emergence of virtual reality, the relevance of nineteenth-century stereoscopic media while also underlining how technological constraints slowed down the development of VR, particularly during the 1980s. Other accounts have focused on how imaginaries of VR concurrently emerged with the rise of Silicon Valley and the countercultural movements of the 1960–80s (Lanier, 2017), or focused on so-called breakthrough devices such as the Oculus headset and the company's troubled path before its acquisition by Facebook in 2014 (Song, 2022).

In this chapter, rather than focusing on the recent development of virtual devices and environments, we will dig into a specific aspect of metaversal histories, namely narrativity. A fundamental presumption of this chapter is that how stories are told and presented to us matters profoundly. Whether backwards-looking or forwards-looking stories have a profound ability to shape how we interpret and give meaning to everything around us as well as ourselves. Indeed, the metaverse itself can be understood as a narrative construction. For instance, Zuckerberg's famous announcement at the Connect 2021 conference was much more about the foundation of a project rather than the presentation of a product. As Zuckerberg underlined during his announcement, the presentation of the metaverse was focused on "how we will build it together" (Meta, 2021) thus positioning the rise of the metaverse in the future rather than in the present.

However, the narrative of the future woven by Meta around the metaverse is not written from scratch; it is the result of a discursive construction that dialogues both explicitly and implicitly with the origins, development, and myths surrounding digital innovations and infrastructures over the past decades. In short, the metaverse imaginary draws on a much larger narrative about technology and/as progress. Metaverse narratives and stories are in themselves infrastructures of reality in that they act not only as forms of intermediation between different temporalities, but also as drivers of scientific and technological experimentation, economic investment, and as justifications for the anticipatory application of technological artefacts, especially those that still only exist on an imaginary level. As Wilhelm Schapp has argued, we are entangled (*verstrickt*) in stories (Schapp 1953); the stories we tell and share about past and future socio-technical developments shape our relationships with technology and the meanings we attribute to it, alongside and beyond our everyday interactions with and through technology. To understand the birth, or perhaps rather the conception of the metaverse, it is essential to trace back its theoretical and imaginary foundations in order to grasp the multiple perspectives that are shaping its current, though still 'magmatic' (Castoriadis, 1987) and uncertain definition.

In what follows below, we retrace and analyse several key narratives that illustrate the recurring fantasy of 'being here and elsewhere' through so-called 'new' media. These media have enabled multiple ways of being—or imaginings of being—here

and elsewhere by holding out hopes for new forms of social, political, and economic interaction and organization in hybrid spaces, not to mention in virtual environments seemingly free from the limitations imposed by the materiality of bodies and things. Being here and elsewhere can mean either being in different places at once or moving rapidly back and forth from one place to another, but it can also mean being able to remain in the same space across different temporalities, inhabiting different bodies and identities. All media technologies tend to challenge our common perceptions of space and time. And each different combination of these technologies has specific characteristics that can stimulate the emergence of different narratives, often utopian or dystopian in character, about our future realities, our spaces, and our bodies with and through such media.

Below, we refer to several key narratives emerging from a plurality of sources, such as sci-fi novels and movies, political speeches and treatises, and corporate communications. In going through these sources, we examine linkages between the history of media and networking technologies with the expected and desirable—and sometimes undesirable—cultural and socio-technical outcomes as foreseen in the 'past futures' (Montross, 2015) or 'imaginary futures' (Barbrook, 2007) of sci-fi novelists, film directors, and other relevant actors from scientific, political and technological milieus.

For the sake of simplicity, this chapter employs a three-step periodization to structure the narratives and related imaginaries found in the history of the metaverse, starting from the first half of the twentieth century. However, the distinctions between these periods are somewhat artificial as elements of each can also be found in others. The first period focuses on the intersection or convergence of various media, such as motion picture, telephony and electricity; the second deals with the combination of digital computing and internet connectivity, while the last two sections focus on the current age and on the specific narrative of the future established by Meta in launching its metaverse project. Overall, this chapter presents a 'short history of metaverse stories', aiming to identify both continuities and changes in the dreams and imaginaries surrounding the metaverse while simultaneously underlining its defining features and sociocultural implications.

2.2 Dream Machines

One of the most famous and foundational myths in media historiography is the so-called 'train effect'. The legend is well known: the first screening of *L'Arrivée d'un train en gare de La Ciotat* by the Lumière brothers in 1896 is supposed to have triggered panic in the audience who believed that the train would crash through the frame, causing them to flee from the theatre. Although this event never really happened, as demonstrated by Bottomore (1999), the myth of the 'train effect' persists as one of the longstanding and most significant in media historiography. The reason is simple. The power of this myth is its symbolic efficiency, that is, it represents both the disruptive power of the cinematographic medium and its ability to radically

change our perception of reality. In *Reality Media*, Bolter et al. (2021) begin their historical account of VR/AR with a second version of this myth. According to other studies on the origins of the train effect, rather than the first screening in 1896, it was the stereoscope experiments conducted by the Lumière brothers in a later screening of the same movie in the 1930s that triggered the awe and panic in the audience and thus gave rise to this myth. This second version emphasizes the important role of early technologies such as stereoscopic lenses, which could lend an illusion of 'greater depth' to images.

Although motion picture is widely acknowledged by media scholars as a radical change in visual culture, some nevertheless argue that stereoscopic cinema was preceded by other spectacular media, such as stereocards:

> Long before both the Lumiere Brothers and the Edison Manufacturing Company filmed their now iconic cinematic actualities of oncoming trains, for instance, many thousands of stereocards were produced that replicated a similar effect of a train emerging from an image towards the viewer (albeit without the movement that later characterized cinema and amazed the viewing public). Nor was this effect limited to trains, with stereographers capturing oncoming ships, zeppelins, artillery, horses, carriages and anything else that would accentuate the stereoscopic effect. (Gurevitch, 2013, 5)

Under the influence of the 'technological sublime' at the beginning of the twentieth century (Marx, 1964; Nye, 1996)—a collective sense of grandeur and wonder engendered by the achievements of the industrial revolution, which appeared capable of uniting nations under the umbrella of technological progress—science fiction writers started to combine elements of this 'new' media system, such as cinema, telegraphy, and photography, with the technological hallmarks of the industrial revolution, such as railway systems and electricity networks. Especially at the beginning of the century, the substitution of face-to-face interactions by other communication forms was a recurrent trope of sci-fi narratives. In the short novel *The Machine Stops* by Forster (1909), for example, future humans are described as living in underground dwellings in which it is possible to accomplish any routine activity, including communicating with thousands of other humans, through a dedicated telecommunication system provided by a god-like machine. In Forster's dystopian narrative, a mother is forced by her son to leave her home in order to take a train and prevent him from abandoning his dwelling. The room where she lives is depicted as a phantasmatic dream—or rather nightmare—of full automation:

> Then she generated the light, and the sight of her room, flooded with radiance and studded with electric buttons, revived her. There were buttons and switches everywhere— buttons to call for food, for music, for clothing. There was the hot-bath button, by pressure of which a basin of (imitation) marble rose out of the floor, filled to the brim with a warm deodorized liquid. There was the cold-bath button. There was the button that produced literature. And there were of course the buttons by which she communicated with her friends. The room, though it contained nothing, was in touch with all that she cared for in the world. (Forster, 1909, 3–4)

This depiction of the mother's room is the result of a pre-digital imaginary in which the mechanization of space can be understood as a precursor to digitalization. The myriads of buttons are a prototypical form of the click, and the dematerialized living

spaces—the bare but hyper-functional room—are a precursor to the empty rooms in which future users live while navigating supposedly abundant virtual worlds with their digital devices. It is interesting to note how sensory elements play a specific role in this story. In Forster's narrative, natural light is frightening. The mother is shocked by the sunlight shining through the train window while travelling, and the main projection of reality in her ordinary life takes place through sounds and other sensory experiences rather than images. The extension or augmentation of reality thus occurs through sound media rather than visual media, which reflects the key role that technologies such as wireless telegraphy and radio were playing before the spread of cinema or television.

A few decades after the publication of Forster's novel, the rise of the motion picture would stimulate new visions and fantasies involving sensory and synesthetic experiences in which the reproduction of visual elements and their interconnection with other sensory phenomena played a crucial role. In 1940, Alfonso Bioy Casares published *The Invention of Morel*, a sci-fi novel that anticipated some of the constitutive elements of the current imaginary of the metaverse. In the novel, the protagonist seeks refuge from a mysterious illness but gets lost on an unknown island inhabited by a group of people who appear to be living in a peculiar, dreamlike state. The protagonist eventually learns that these people are not quite real but are rather the result of an experiment involving a machine that can recreate human beings and their lived experiences, capturing them in a kind of eternal loop. This loop is a recording of the everyday life of the inventor, Morel, and his friends and fellows; a machinic dream created to keep alive the memory, and especially the feeling, of his romantic relationship with a woman named Faustine. In a note addressed to future visitors of the island, Morel describes his machine as follows:

> With my machine a person or an animal or a thing is like the station that broadcasts the concert you hear on the radio. If you turn the dial for the olfactory waves, you will smell the jasmine perfume on Madeleine's throat, without seeing her. By turning the dial of the tactile waves, you will be able to stroke her soft, invisible hair and learn, like the blind, to know things by your hands. But if you turn all the dials at once, Madeleine will be reproduced completely, and she will appear exactly as she is; you must not forget that I am speaking of images extracted from mirrors, with the sounds, tactile sensations, flavours, odours, temperatures, all synchronized perfectly. An observer will not realize that they are images. And if our images were to appear now, you yourselves would not believe me. Instead, you would find it easier to think that I had engaged a group of actors, improbable doubles for each of you! (Bioy Casares, 1940, 88)

Morel's machine works thanks to a perpetual energy source powered by sunlight and sea tides; its magic lies first of all in reproducing reality due to the combination of the cinematographic medium with an unprecedented 'green' energy source. Morel's dream is a meta-reality that brings with it immortality, and in particular, the immortalization—which is both the perpetuation and the immobilization—of a social bond (in this case love) through a permanent projection of its lived experience. The island, the infinite engine, and the projection apparatus are a socio-technical assemblage representing a long-standing fantasy behind simulation technologies: overcoming our biological constraints (Davis, 2015). Another relevant classic of

early twentieth-century sci-fi literature, Aldous Huxley's *Brave New World* (1932), similarly contains some important precursors to the synesthetic motion picture, but with more pessimistic and dystopic nuances compared to Bioy Casares. In Huxley's novel, one of the main forms of collective entertainment is a special type of movie theatre called a 'feelie' in which films can stimulate all the audience's senses. A feelie merges visual perception with tactile and olfactory sensations, enabling viewers to physically experience the emotions and sensations portrayed in the movie (feelie is a portmanteau of the terms feeling and movie). For Huxley, however feelies represent a form of entertainment in which real emotions, relationships, and even high art are replaced by superficial experiences, thus sacrificing individuality and authenticity in favour of new technocratic forms of control:

> That's the price we have to pay for stability. You've got to choose between happiness and what people used to call high art. We've sacrificed high art. We have the feelies and the scent organ instead. (Huxley, 1932, 151)

The dystopian narrative presented in Huxley's novel coincides with the critical theory of communication that was contemporaneously emerging among scholars from the Frankfurt School, especially through the work of Adorno and Horkheimer (1947). These critics of the culture industry, which they considered a driving force of so-called 'soft totalitarianism'—saw cinema, as well as all forms of broadcasting media, as the main weapon of mass distraction used to enculturate or indoctrinate people into capitalism and consumerism. The role of media in establishing and maintaining cultural hegemony are longstanding topics that have been recently taken up by critical scholars such as Žižek (2000) and Fuchs (2020), among others. Even more recent theories, such as Cory Doctorow's concept of "enshittification" (2025)—i.e. the planned, gradual deterioration of online platforms, services, and products—go in a similar direction, underscoring the false sense of freedom associated with consumers' choices and the concentration of power in the hands of 'Big Tech' companies and their marketing and profiling strategies.

Science fiction literature from the first half of the twentieth century thus offers an interesting array of visionary narratives about the future of media technologies, laying the foundation for several recurring topics as well as critical perspectives on the relationship between media, reality, and society. However, in all these past futures, a genuine convergence between communication technologies and multisensory experiences, essential for the current conceptualization of the metaverse, was not yet being contemplated. In a similar vein, other cultural works, such as Fritz Lang's *Metropolis* (1927), acted as a bridge between the ancient myth of Prometheus, Mary Shelley's *Frankenstein*, and the rise of artificial intelligence, thus anticipating some key themes in the later emergence of AI. One pertinent example of this is the hubris of scientists seeking to create a being as intelligent as humans, if not superior to them, thus stealing from God the act of creation just as Prometheus had stolen fire from the Greek gods. However, it was only in the 1960s, concurrently with the rise of digital computing and computer networks, that ideas about connectivity and the dematerialization of experience began to be combined in Western sci-fi narratives as well as the broader sociotechnical imagination.

2.3 Digital Infrastructures as Stories

The biggest obstacle to the realization of the metaverse, particularly as conceived by Meta today, is probably the so-called interoperability issue, that is, the creation of a network of interconnected worlds through which one can navigate freely without the limits imposed by different and incompatible standards, devices, and operating systems. In the history of digital media and infrastructures, similar concerns about issues of interoperability between different computer systems emerged immediately after World War II, but these were fully addressed only decades later with the spread of the internet and especially with the birth of the World Wide Web in the mid-1990s.

Between the late 1940s and the 1970s, the great popularity of science fiction—think of authors such as Isaac Asimov, Philip K. Dick, and Kurt Vonnegut—went hand in hand with the proliferation of futurist narratives spread by computer scientists and cyberneticians, who, during the first phases of the digital revolution (Balbi, 2024) started to conceptualize what relations and interactions between humans, computers, and information infrastructures might look like (Bory, 2020). From Vannevar Bush's Memex (1945), Joseph Licklider's intergalactic network (1960), Douglas Engelbart's augmented intelligence (1962), and the hypertextual dream of Xanadu by Ted Nelson (1967) to the subsequent development of hypertext, Arpanet, and then of the internet, all these developments took place alongside the formulation of a wide variety futuristic scenarios. More precisely, in these decades, the challenge of achieving interoperability between communication systems fed into at least two diverging narratives concerning freedom in the digital future. On the one hand, computer networks were conceptualized as technologies that would lead to new forms of collective intelligence and knowledge sharing, and in the most extreme version of this narrative, to a complete separation between the 'real' and a new and better digital world. On the other hand, in another narrative, created by Western governments and later reproduced during the Al Gore vice-presidency in the 1990s, the internet and the Web were portrayed as new highways that would connect a global market in a world free from war and geopolitical conflicts. This second narrative also influenced several local political movements, such as civic networks, as we discuss further below.

The first narrative is symbolized by the concept of cyberspace, first introduced in the cyberpunk novel *Neuromancer* by Gibson in 1984. In the 1990s, cyberspace was further conceptualized as an entirely new interconnected system that could serve as the infrastructure for a new information universe. This reformulation, however, opened up a series of utopian and dystopian scenarios. For example, the perception that humanity might be able to transcend the material constraints of geography, politics, and corporeality led to suggestions for the creation of a so-called collective intelligence that would surpass that of the single units of which it consisted (Lévy, 1997) as well as to anarchic ideas about a self-regulated virtual world in which everyone would be free and able to express themselves as they might wish. However, such dreams of collective intelligence or a society freed from political and economic constraints would soon be crushed by the rise of surveillance capitalism, which would make data and interoperable infrastructures its main source of value.

The culmination of the narratives surrounding cyberspace in the 1990s—a space freed from space—is the famous *A Declaration of the Independence of Cyberspace* shared on the Web by John Perry Barlow during the World Economic Forum at Davos in 1996:

> Governments of the Industrial World, you weary giants of flesh and steel, I come from Cyberspace, the new home of Mind. On behalf of the future, I ask you of the past to leave us alone. You are not welcome among us. You have no sovereignty where we gather. […] Cyberspace consists of transactions, relationships, and thought itself, arrayed like a standing wave in the web of our communications. Ours is a world that is both everywhere and nowhere, but it is not where bodies live. We are creating a world that all may enter without privilege or prejudice accorded by race, economic power, military force, or station of birth. We are creating a world where anyone, anywhere may express his or her beliefs, no matter how singular, without fear of being coerced into silence or conformity. Your legal concepts of property, expression, identity, movement, and context do not apply to us. They are all based on matter, and there is no matter here. (Barlow, 1996)

The rhetorical force of Barlow's *Declaration* lies in its combination of specific adjectives and metaphors. On the one hand, the giants made of flesh and steel, on the other, an immaterial mind independent of physical constraints. In cyberspace, according to Barlow, where there is no matter and where reality is "a standing wave in the web of communications", there is no material power either.

Barlow's declaration was one of the first texts to go viral on the Web. Importantly, the rhetorical and material dimensions of this declaration were deeply intertwined. Until the late 1990s, networking infrastructures could handle only small amounts of data. The dematerialization of geographical distance could mainly be achieved through a so-called light and poor medium, such as text, rather than a more data-heavy medium such as image or sound. These limitations in terms of bandwidth, as noted by virtual-reality pioneer Jaron Lanier (2017), contributed to the slowdown of heavier technologies, such as VR, which would consequently remain in the realm of sci-fi narratives and video arcades.

As a consequence, text was fundamental to the development of early cyberspace and virtual identities. In particular, beside developing their avatars, early users created and fostered new online identities in chatrooms and early MMPORGs (Massively multiplayer online role-playing games), literally writing themselves down via instant messages or asynchronous conversations with others and thus weaving their relationships with these so-called second selves (Turkle, 2005). According to Turkle, in the digital age technology became a new "architect of our intimacies" (Turkle, 2005). While the first avatars allowed for early explorations in reshaping identity, it was online relationships that were seen as a means for exploring the self in a more authentic manner than compared to 'real' physical interactions. During this period, the online creation of avatars and fictional characters was seen as a way to project desires about one's body, gender, and social identity (Casilli, 2010). In cyberspace one could embody not only someone else – if not something else, such as a monster or a fantastical creature in roleplaying games – but also explore more authentic selves in ways far removed from the restrictions imposed by social conventions or expectations and stigmas. These new digital spaces had a strong playful connotation and,

thanks to anonymity and the lack of institutional and jurisdictional oversight, were perceived as neutral environments that were less likely to involve social judgment or have reputational repercussions on the 'real' social life of users. However, enthusiasm for these new mediated socialities and experimental identities was accompanied by fear about individual and collective loss, particularly in relation to the so-called myth of immersivity according to which virtual spaces risked turning into cages from which individuals could no longer escape, or would no longer want to escape. This fear of being enclosed has ancient origins, starting from Plato's cave to the total cinema (Bazin, 2004) and reappears in the harsh critiques of simulation and virtuality formulated by French theorists such as Baudrillard (1981) and Virilio (2000), among others.

Curiosity and uncertainty about the nature of virtual practices and the future of the digital society led to the production of a wealth of sci-fi movies between the 1980s and 1990s—a golden age for the genre of sci-fi cinema—dedicated to exploring the relationship between the body, memory, identity, technology, and the emergence of augmented and virtual reality devices and environments. From the video game immersion of Steven Lisberger's *Tron* (1982) to the cyberpunk visions of the console-body in David Cronenberg's *Existenz* (1999) and the relivable memories of the SQUID (Superconducting QUantum Interference Device) in Kathryn Bigelow's *Strange Days* (1995), up to the scientistic hubris of Brett Leonard's *The Lawnmower Man* (1992) and the grotesque virtual game character of Gabriele Salvatores' *Nirvana* (1997), cinematic science fiction over these two decades visually represented and interrogated the various fears and pleasures that digital immersion, including diving into our own brains and memories, might produce. Such movies and the imaginaries of virtual reality they contained have already received much scholarly attention (e.g. Chan, 2014). Some lesser-known cinematic productions, however, have explored other interesting aspects of virtual immersion, particularly from a sociological point of view. For example, *Thomas in Love* by Pierre Paul Renders (2000) in which the protagonist, a thirty-year-old man named Thomas, has never left his home since childhood. Like the mother in Forster's novel, Thomas's life is in a state of permanent connection in which the screen has replaced Forster's buttons; he goes to the doctor, orders food, and does everything he needs to live online while sitting in his dark room. His agoraphobia makes Thomas incapable of dealing with the outside world or having any human contact, his life is on the internet. Films such as *Thomas in Love* were released at the turn of the millennium when the diffusion curve of the internet and personal computing was rising globally. It is at this juncture that the Wachowski siblings released their movie *The Matrix* (1999), which combined dystopian aspects of the immersion narrative with ideas about the advent of so-called strong AI, and subsequently resulted in one of the most successful narratives in the history of sci-fi cinema. *The Matrix* is so far the only imaginary world in which the concept of framelessness is realized and represented (Carter & Egliston, 2024), as the main medium between the human characters and the virtual world they live in is located in their brain; there is no means for the characters to distinguish between what is real or simulated.

One year before the release of *The Matrix*, another movie dealing with an alternative form of reality attracted the attention of the general public as well as that of critical scholarship. Relying on the 'old' medium of television, *The Truman Show* (1998) by Weir contrasted the dystopia of virtual reality as represented by *The Matrix* with a more plausibly realistic but equally fictional setting in which reality is mediated by human actors rather than by AI-controlled simulations. The protagonist Truman ('the only true man') unwittingly lives in a Big Brother-type show where all the other people are actors playing their part and co-running the television show that has been broadcasting his entire life from birth. *The Truman Show* forcefully challenges the idea that virtual reality and fictional lives must be mediated by technology. We will return to this point later, after exploring the emergence of another narrative in the 1990s—arising concomitantly with the cyberspace narrative—which connects the realm of the virtual with civic participation rather than individual immersion.

2.4 Netizens and Early Digital Twins

The conceptualization of the internet as a free and anonymous space for experimenting with the self and for recreating new forms of virtual sociality was accompanied, especially during the late 1980s and early 1990s, by the emergence of another narrative in which the virtual realm was imagined as an enhancement of the public sphere rather than as its replacement with an unregulated cyberspace. While Western countries were investing huge material and symbolic resources in promoting the so-called information superhighways, the internet and especially the Web seemed to provide the foundation for a new democratic arena in which citizens could directly partake in political decisions. This chapter is not the place to dig into the history of the transformation of these so-called virtual communities (Rheingold, 1993) and their role in the rise of the so-called network society (Castells, 2000). However, it is important to highlight the narratives of civic participation that emerged in parallel to the alleged dematerialization of digital space. This is particularly the case because the same digital tools that were once imagined as boosters of grassroots activism and collective participation in politics—at first internet connectivity but later also smartphones and social media—are today often criticized as the causes of the people's detachment from the *res publica* rather than enablers of collective politics.

In the 1990s, imaginaries of a new active netizenship (Hauben and Hauben, 1997) were emerging, which were posited, with variable degrees of success, as important alternatives to the framing of the cyberspace as a virtual world separated from concrete reality. It is in this context that the first civic networks were born as platforms of intermediation and co-participation between national governments, local municipalities, and citizens, especially in several major European cities. One of the first civic networks was De Digitale Stad (DDS), launched in 1994 in Amsterdam (Alberts et al., 2017), but other important civic initiatives, such as Bologna's network Iperbole, created in 1993, were also set up to provide free internet access to citizens for the first time in Europe (Bory, 2019). Such experiments in fostering active citizenship through

the internet involved different stakeholders, such as activists, local governments, and prominent philosophers and computer scientists, as well as common citizens who contributed to the construction of a new online public sphere (the 'netsphere'). These civic initiatives can be compared to the emergence of previous grassroots initiatives, such as pirate radio stations and early computer hobbyist communities. Notably, these European civic networks combined multiple visual and textual elements that anticipated the birth of the so-called digital twins (see next chapter), such as navigable maps of cities as well as virtual squares where users could access a wide range of services, read news items, and find practical information about their municipalities' activities and policies (Tambini, 1998). These services were made available alongside online discussion groups in which citizens could collectively debate policies and thus participate in decision-making processes, exchange goods and professional services, and provide feedback on the efficacy of actions taken by governance institutions at local and regional levels.

Amsterdam's DDS has been preserved in the UNESCO digital heritage platform and is described as follows:

> On January 15, 1994, De Digitale Stad (DDS, The Digital City) opened its virtual gates. It empowered users to be "citizens" or "netizens" of a digital environment and enter the then largely unknown world of the Internet. DDS imagined a community-driven, commons-based internet, far remote from today's internet whose operation is dictated by a small group of big tech companies. (UNESCO, 2023)

DDS employed the architectural metaphor of 'town squares' to guide citizens in finding information on the city's website and included gamification features that allowed users to configure their own, quite basic, avatars that could be used to navigate the website and participate in online activities. Here, in contrast to the cyberspace narrative in which the online 'world' was seen as detached from real world politics, the internet was conceptualized by civic network movements as a new space for enhancing public participation and for recreating the publicness and openness of 'real life' spaces. Practices of online "square-hopping" (Francissen & Brants, 1998) fostered a feeling of co-presence and co-participation among citizens in this new virtual square, and, moreover, also encouraged face-to-face meetings and encounters in physical spaces. Unlike social media platforms—defined by the critical scholar Geert Lovink, one of the founders of DDS, as "networks without a cause" (Lovink, 2011)—civic networks were created and narrated as tools for re-building and re-enacting direct democracy. These civic networks also served as precursors to a series of later and larger-scale political movements (such as the 5 Star Movement in Italy and the Pirate Parties in Northern and Eastern Europe) that would make the network-citizenship bond a core point of their political and organizational identity. While Neil Stephenson wrote about a future metaverse monopolized by 'Big Tech' companies and dominated by economic elites, in the 1990s the internet was still seen as a tool to speed up direct democracy as well as an infrastructure to facilitate other activities such as teleworking and telemedicine.

However, network technology of the 1990s was not sufficiently advanced for the realization of such dreams of democratic togetherness or of a truly liveable virtuality. These virtual worlds were still relatively poor in terms of internet connectivity,

graphics, and technological sophistication. Nevertheless, they were also rich in the sense that they embodied and inspired collective imaginations and expectations of the future. Soon, however, these political ideals and expectations would be betrayed, only to be taken up again a few decades later, yet this time shaped by a different vision of a sociotechnical future espoused by one particular leading company.

2.5 Birth and Death of the Metaverse

The speed with which computing power and internet connection have developed from the late 1990s to today has facilitated the spread of increasingly sophisticated and complex technologies, software, and virtual environments, but these, however, require a long process of domestication on the part of users that is not only technical but also symbolical, physical, and psychological. Although, as Carter and Egliston have aptly shown (2024, 5–28), companies such as Sega and Nintendo failed several times in their attempts to make VR a global phenomenon, this is to a great extent due to psycho-physical effects such as motion sickness as well as users' hesitancy to be fully immersed in multisensorial virtual landscapes and experiences. After all, calibrating bodily senses and technological standards at the collective level is a delicate and slow process. For example, it took decades to make 3D visuals familiar and user-friendly, even for enthusiastic expert users such as gamers, via first-person games played on conventional screens. Only recently has this process of familiarization and habituation started for mixed-reality headsets. From a technological perspective the history of the metaverse is part of the long history of visual media, ranging from pictorial art to 3D virtual environments, and part of a process in which technologies of various kinds transformed from the realm of the sublime to the banal (Mosco, 2005) over the span of decades, if not centuries.

The history of the metaverse also involves the decline and (false) resurrection of a narrative of togetherness (Simonson, 1996) that posits media technologies as revolutionary and democratic infrastructures. Notably, the beginning of the twenty-first century marked the end of both the utopian narrative of cyberspace and the democratic narrative of digital citizenship. Both myths or sublime visions (Mosco, 2005), once subjected to scalability by 'Big Tech' companies, have succumbed to the dynamics of digital capitalism. The "net delusion" (Morozov, 2011) of the Arab Spring movements, as well as the rise and fall of the "people of the net" (Natale & Ballatore, 2014)—both once hailed as important emerging political movements—are striking examples of how internet technology has failed to replace the infrastructures of political intermediation. But above all, these developments demonstrate how digital infrastructures and media—as well as the narratives surrounding them—have not been able to replace the role that grand narratives and political ideologies have played in holding societies together. For decades, several cyber-communitarian movements—most of them from the left of the political spectrum—regarded technologies such as the internet and computing as vehicles for advancing egalitarian and

democratic processes. More recently, however, new systems and technologies, such as the metaverse and AI, have come to represent civilizational progress in themselves.

Interestingly, one of the main architects behind the demolition of the myth of cyberspace was Mark Zuckerberg. In 2004, Linden Lab's Second Life seemed to open the doors to the creation of a truly interconnected virtual world. The growth of computational power and internet connectivity gave impetus to a new virtual realm whose slogan was, "Your world, your imagination", which triggered the interest of social scientists and ethnographers trying to understand this new environment and form of sociality consisting of multicoloured landscapes and outlandish avatars in which U2 played its first virtual concert and where users were creating and shaping their second lives. We will not elaborate here on the main features of this well-known virtual world, which, despite criticism, re-emerges in some narratives as the forerunner to if not the only true metaverse (Au, 2023). What is interesting to note from a historical and narratological point of view is that the decrease in interest in Second Life during the late 2000s coincided with the global diffusion of social media platforms, especially with the birth of Facebook and particularly its rise from 2007 onwards. What Facebook's slogan of making the "world more open and connected" (Hoffman et al., 2018) made clear—together with similar slogans such as YouTube's "broadcast yourself"—was that the greatest source of inspiration (and financial profit) in the digital age was not fantasy or imagination but reality. Rather than imaginary worlds, creative avatars, or new social relations, the emergence of social media and 'reality' television placed self-broadcasting and ordinary lives at the centre of the internet as well as the broader media landscape.

There are many factors lying behind the paradigm shift that affected the entire digital landscape, and one of them is certainly financial. Frankly, big tech companies realized that real selves were much more profitable than imaginary ones. Our faces, clothes, voices, tastes, and choices have become the main resource of the digital economy in the form of data. Starting from the birth of the Google search engine in the late-1990s, through to the social media platforms and the emergence of Web 2.0 in the 2000s and the current rise of generative AI, personal data has been front and centre. The data 'Big Tech' companies collected over time on users and their networks, also thanks to the broadcasted selves mentioned just above, have become a financial gold mine and a fundamental resource for consolidating and perpetuating their dominant position in the global market. However, these companies nevertheless still have to cultivate and maintain their symbolic power in addition to their economic and technological powers. 'Big Tech' companies must appear as cutting-edge and ahead of the times in order to maintain the interest of investors. Corporate rebranding strategies point exactly in this direction; companies such as Facebook and Google change their names, make promises of progress and radical technological change, and pretend to open the market doors to new players and competitors. But their main goal is maintaining the current power (un)balance and their dominant position and control over the digital market and the data economy. Historically, powerful actors have constantly needed to demonstrate their ability to adapt to the 'new', if not just to anticipate and shape the conditions for progress and change; contemporary corporate giants are no exception to this.

2.6 2021—The Resurrection

Digital media history is rich in mythopoetic sources such as governmental documents, academic papers, manifestos, and letters from CEOs to investors, among other examples. Some of these documents have been chronicled as key turning points in history. Think of Alan Turing's seminal paper (1950) that proposed the famous Turing Test and how this started a never-ending debate on the meaning of 'artificial intelligence', or the aforementioned *A Declaration of the Independence of Cyberspace* by Barlow (1996) and its juxtaposition of national states with anarchic networks, or how the *Computerization of Society* by Nora and Minc (1978) fed into European and US discourse on the need for digitization and high-speed infrastructures in the 1980s and 1990s, or how the famous *An Open Letter to Hobbyists* written by Gates (1976) heralded in the age of proprietary software and eventually led to the creation of the neoliberal software marketplace.

In relation to such canonical texts, and well before the launch of Meta in 2021, Mark Zuckerberg had already employed the medium of the letter to glorify and narrate the past and the future of his company as well as its ostensibly key role in society at large. In 2012, the CEO of Facebook wrote a famous letter to investors that sought to gain their trust by underscoring the company's outstanding accomplishments as well as its contemporary and future achievements:

> At Facebook, we're inspired by technologies that have revolutionized how people spread and consume information. We often talk about inventions like the printing press and the television - by simply making communication more efficient, they led to a complete transformation of many important parts of society. They gave more people a voice. They encouraged progress. They changed the way society was organized. They brought us closer together. [...] Today, our society has reached another tipping point. We live at a moment when the majority of people in the world have access to the internet or mobile phones - the raw tools necessary to start sharing what they're thinking, feeling and doing with whomever they want. Facebook aspires to build the services that give people the power to share and help them once again transform many of our core institutions and industries. (Zuckerberg, 2012)

Inspired by other prominent 'tech' CEOs such as Steve Jobs, Zuckerberg placed his company in a direct line with the historical development of other media, particularly when claiming that Facebook was the networking infrastructure that had wired, and would rewire, social, economic and cultural connections worldwide, just as previous revolutionary technologies such as the printing press and television had "led to a complete transformation of many important parts of society". This narrative fits perfectly with the concept of corporational determinism (Natale et al., 2019), that is, the recurring strategy of digital media corporations to narrate themselves as the main agents of societal and technological change. The appropriation of the imaginary of other characters and communities in 'tech' and media history is also illustrated by the subtitle of Zuckerberg's letter, "The hacker way", which suggests that the company embodied the values of hacker culture. However, the hacker movement has long been critical of closed systems such as Facebook's platforms as well as companies employing end-to-end processes such as Apple.

Almost ten years later, in his letter of October 2021, Zuckerberg appeared to reenact a similar narrative:

> We are at the beginning of the next chapter for the internet, and it's the next chapter for our company too. [...]The metaverse will not be created by one company. It will be built by creators and developers making new experiences and digital items that are interoperable and unlock a massively larger creative economy than the one constrained by today's platforms and their policies. [...] To reflect who we are and the future we hope to build, I'm proud to share that our company is now Meta. Our mission remains the same — it's still about bringing people together. Our apps and their brands aren't changing either. We're still the company that designs technology around people. But all of our products, including our apps, now share a new vision: to help bring the metaverse to life. And now we have a name that reflects the breadth of what we do. [...] Ours is a story that started in a dorm room and grew beyond anything we imagined. […] We have built things that have brought people together in new ways. We've learned from struggling with difficult social issues and living under closed platforms. Now it is time to take everything we've learned and help build the next chapter. [...] I'm dedicating our energy to this — more than any other company in the world. If this is the future you want to see, I hope you'll join us. The future is going to be beyond anything we can imagine. (Meta, 2021)

Read through an historian's lens, Zuckerberg's letter looks like a hodgepodge of the narratives and the past futures presented and analysed in this chapter. Perhaps the most contradictory aspect of his letter is that the end of the platform economy and the emergence of a new interoperable and horizontal system—the metaverse—will supposedly be driven by one of its leading players, if not the archetype of the platformization of the internet (Bucher, 2021; Plantin et al., 2018). Moreover, situating the humble origins of the great company in a college dorm room reiterates an old cliché of Silicon Valley, namely that of the genius who emerges from below, pushing his revolutionary discovery in the service of the entire society, but ignores the fact that a dorm room at Harvard is not that low on the social scale of the wealthiest country in the world. Furthermore, Zuckerberg's Metaverse is very similar to the initiative that he contributed to undermining by founding Facebook in the mid-2000s—i.e. Second Life and virtual worlds—but his version is emptied of the ideals that such metaverses once embodied, such as the search for new forms of community and self-exploration that surrounded virtual-reality technologies and digital worlds of the 1990s and early 2000s. From the point of view of collective imaginaries, the resurrection of the metaverse by Meta is characterised by a degree of degradation that is akin to the zombies found in fantastical literature; creatures resurrected by the force of technology but that have lost their humanness in the process. Rather than the sci-fi imaginary presented by Stephenson in *Snow Crash*, which challenges the idea of a horizontal and 'democratic' metaverse—or the rhetoric of technological and perceptual revolution surrounding VR technologies—Meta's narrative of the Metaverse is buttressed only by its economic power. The ten thousand jobs in Europe promised by Meta for the development of metaverse technologies, its influence over the news media as one of the largest global platforms, the acquisition of Oculus and the dominance of the Quest headset in VR hardware, and its ownership of the most widely used social media platforms and messaging apps in the Western world; these are the reasons that have made the metaverse and its entanglement with Meta

one the main topics of public, political, economic and academic debate, this book included. Even though major Hollywood productions such as *Avatar* (Cameron, 2009) and *Ready Player One* (Spielberg, 2018) reproduced dominant imaginaries surrounding immersive technologies, such narratives were more inspired either by questions around the existence of other species, such as in the case of *Avatar*, or by interest in exploring the role of public figures and cultural references from the last decades of the twentieth century. In *Ready Player One*, for example, the character of the proto-Steve Jobs and nostalgia for role-playing games of the 1980s–1990s functioned more as representations of the "future that used to be" (Barbrook, 2007) rather than of the future imagined today by companies such as Meta.

We should not deny that it is possible that virtual technologies and environments are rapidly evolving and could well become part of our daily lives in the short or long term, in different ways, and potentially with more or less desirable outcomes. Just as it is necessary to examine narratives of the metaverse imposed from above by 'Big Tech' companies, it is equally necessary to highlight other visions of the metaverse's potential future as well as to uncover missing perspectives that mainstream narratives can conceal.

An emergent body of scholarship argues that hype is replacing hope in contemporary imaginaries (Galanos & Stewart, 2024). If we consider hope exclusively as a collective phenomenon aimed at the common good, it is probably true that technological hype is now supplanting the dreams of togetherness that were associated with media of the past such as radio, television, and the internet. But hope provides nourishment to ideals of progress found in democratic societies as well as in companies. The 'launch' of the metaverse, however, represents specific corporate hopes and imaginaries in which embodied experience and virtual environments are tools to sustain the delusions of grandeur of a small group of actors rather than tools to support broader democratic ideals or provide to social needs.

During an interview right after the launch of Meta in 2021, Zuckerberg claimed that the metaverse is "an embodied internet and you're in it and *the atomic unit is you* have your avatar and your digital goods. […] I think that's an architecture that should be fundamentally more amenable to interoperability" (Heath, 2021, emphasis added).

This short excerpt highlights how single users are conceptualized as "the atomic unit" of a metaverse in which individuals, and particularly their bodies, are part of the "architecture" of the virtual world envisioned by Meta. More than one century ago, however, understanding and mitigating the atomization of society was considered one of the foremost challenges of modernity, an issue that led to the birth to the social sciences and that stimulated seminal reflections by authors such as Ferdinand Tönnies and Georg Simmel on the fragmentation of society and the decline of social reciprocity. Zuckerberg's imaginary of a metaverse—consisting of billions of connected monadic elements linked "alone together" (Turkle, 2011), using their datafied bodies to work and play in quantified spaces—brings this genealogical question back to the heart of the social sciences and humanities. It is precisely for this reason that hidden or alternative narratives and even counter-narratives should be

identified and highlighted; to understand what the metaverse is or can be, but also what it will do to us, our bodies, and our social and living spaces.

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